

ANDY ALEXANDER

Senior Linux Systems Engineer

Ormond Beach, FL (open to remote)

github.com/greenbeanorg | linkedin.com/in/andy-alexander-linux

SUMMARY

Senior Linux systems engineer with 15+ years of hands-on production experience spanning Linux, virtualization, networking, storage, web platforms, monitoring, and automation. Enterprise background includes high-volume transactional email infrastructure at Sendwell and globally distributed web infrastructure for ESPN.com and ESPN3.com at The Walt Disney Company, covering VMware/Proxmox virtualization, load balancing, configuration management, and multi-layer production troubleshooting. Since 2022, has independently designed and operated multi-site Linux infrastructure — Proxmox, ZFS, Docker, routing and firewall policy, VPN, monitoring, and backup/recovery — documented as version-controlled operational runbooks.

TECHNICAL SKILLS

Linux & Systems: Linux — Red Hat Enterprise Linux (RHEL), CentOS, Debian, Ubuntu; systemd, Bash, performance troubleshooting, system hardening, service lifecycle management, FreeBSD, Windows Server

Infrastructure & Virtualization: Proxmox VE, VMware ESXi, KVM, LXC, Docker & Docker Compose, TrueNAS SCALE, high availability, capacity planning, incident response, root-cause analysis, lifecycle management, PCIe/SATA passthrough

Automation & Configuration Management: Bash, Git/GitHub, API-driven automation, declarative configuration, systemd timers and cron, operational scripting, SaltStack, AI-assisted Python/Node.js scripting with human review and testing

Networking: TCP/IP, VLANs, IP subnetting, DNS/DHCP, routing, firewalls, NAT/nftables, WireGuard, OPNsense, MikroTik RouterOS, QoS/traffic shaping, tcpdump, Cisco IOS, F5 BIG-IP, OpenVPN, OSPF/MPLS troubleshooting exposure, Citrix NetScaler

Storage & Backup: ZFS/RAIDZ, NFS, LVM, restic, automated backup and recovery, retention management, integrity verification, SMB, mdadm RAID

Monitoring & Observability: Grafana, ELK/Elastic Stack, Uptime Kuma, Nagios, syslog/journalctl, alerting and escalation, performance monitoring, operational runbooks, NUT, Cacti

Cloud: Oracle Cloud Infrastructure — Ampere A1 ARM64 compute, VCN networking, security lists, public-facing and off-site services

Web, Databases & Applications: Nginx, Apache, PostgreSQL, MySQL, high-volume MTA infrastructure, JBoss, GlassFish

Security: Linux hardening, SELinux/AppArmor, nftables/iptables, Fail2ban, PCI compliance, Nmap, Nessus, OpenBSD pf

PROFESSIONAL EXPERIENCE

Infrastructure Engineer — Independent

2022 – Present | Ormond Beach, FL

Design, build, operate, and document a multi-site Linux environment spanning virtualization, storage, networking, monitoring, backup/recovery, and containerized services. Architecture and operational procedures maintained as version-controlled documentation at github.com/greenbeanorg.

- Designed and operate a multi-host Proxmox environment running Linux services, network infrastructure, storage, home automation, and application workloads across physical and virtual hosts.
- Migrated approximately 30TB of storage from mdadm RAID5 to TrueNAS SCALE with a ZFS RAIDZ1 pool using PCIe SATA passthrough under Proxmox; redesigned cross-host storage access around NFS.
- Built automated restic backup pipelines across three Proxmox nodes, firewall infrastructure, and home automation, with 90-day retention, scheduled integrity verification, and off-site replication to Oracle Cloud Infrastructure.
- Built declarative monitoring automation on Uptime Kuma, with monitor definitions maintained as inventory and reconciled through API-driven provisioning across virtualization, storage, networking, DNS, and application services.
- Deployed and operate multi-container Docker Compose application infrastructure, resolving startup dependencies, UID/GID permission mapping, container networking, and hardlink-aware ZFS dataset layout.
- Designed and deployed a WireGuard hub-and-spoke VPN connecting two sites and roaming clients through a cloud-hosted hub; implemented subnet routing and asymmetric NAT policy, and diagnosed tunnel bring-up failures using packet capture.
- Operate OPNsense routing and firewall infrastructure with MikroTik switching; engineered WAN traffic shaping with pipes and queues to control bufferbloat, holding an A+ bufferbloat grade at ~361/317 Mbps with +0/+2 ms added latency under load.
- Eliminated recurring ISP outages caused by a gateway firmware memory leak by replacing the vendor device with an XGS-PON ONT-on-a-stick SFP+ running 8311 community firmware; flashed the module, set OMCI identity via U-Boot variables, and validated PLOAM association and stable service.
- Migrated an ARM single-board host from an orphaned vendor kernel to a maintained Armbian/Ubuntu stack after analyzing U-Boot boot-partition dependencies; deployed redundant Pi-hole DNS across independent failure domains, and standardized UPS monitoring on NUT in netserver/netclient architecture.

- Author structured operational runbooks documenting architecture, design decisions, failure modes, recovery procedures, and verification steps, maintained under Git version control.

Environment: Linux, Proxmox VE, TrueNAS SCALE, ZFS, Docker Compose, OPNsense, MikroTik RouterOS, WireGuard, Armbian, restic, NUT, Uptime Kuma, Bash, Git, Oracle Cloud Infrastructure.

Contract Systems Engineer — Sendwell, Inc.

2015 – 2022 | Bloomfield Hills, MI / Remote

- Provided intermittent, project-based infrastructure engineering and modernization for a high-volume email marketing platform, including production troubleshooting, upgrades, migrations, and infrastructure changes.

Senior Systems Engineer — Sendwell, Inc.

2013 – 2015 | Bloomfield Hills, MI

- Designed and operated high-volume transactional email infrastructure on Linux (Ubuntu/Debian/CentOS), VMware ESXi, and Proxmox, supporting thousands of concurrent transactions and sustained network throughput of approximately 5 Gbps.
- Operated clustered mail infrastructure, 10GbE switching with static upstream routing, PostgreSQL services, and 40Gb InfiniBand-connected storage.
- Built custom Nagios and Cacti monitoring covering more than 200 VMs across the physical host fleet, and automated backup workflows using rsync, shell scripting, and ghettoVCB against the ESXi CLI.
- Implemented centralized ELK logging with Kibana dashboards for Apache/Nginx web infrastructure, enabling analysis of request traffic, source activity, and HTTP response patterns.
- Managed core infrastructure services including OpenLDAP, DHCP, DNS/BIND, and office wireless infrastructure.
- Maintained SaltStack configuration management and internal MediaWiki and RequestTracker infrastructure.
- Troubleshot complex production incidents spanning operating system, network, storage, virtualization, and application layers.

Environment: Debian, Ubuntu, CentOS, VMware ESXi, Proxmox, PostgreSQL, OpenLDAP, BIND, InfiniBand, Nagios, Cacti, ELK, SaltStack, Bash.

Senior Systems Engineer (contract) — ND Industries, Inc.

2012 | Troy, MI

- Led a mixed Linux/Windows Epicor ERP implementation and production rollout with emphasis on infrastructure readiness, monitoring, and operational documentation.
- Implemented Nagios monitoring and authored comprehensive internal system procedures and documentation.

Senior Systems Engineer — The Walt Disney Company (ESPN)

2011 | Seattle, WA

- Supported application, load-balancing, and web infrastructure for ESPN.com and ESPN3.com using F5 BIG-IP and Red Hat clusters.
- Completed a live data center migration of ESPN.com, ESPN3.com, and subsidiary properties.
- Troubleshot production services under peak national traffic, including high-visibility ESPN.com services during NFL Sundays.
- Administered and clustered Sun GlassFish Java application servers and performed capacity planning using SolarWinds.
- Automated system configuration and enforced fleet consistency using CFEngine3.

Senior Systems Administrator — Entertainment Publications

2009 – 2011 | Troy, MI

- Maintained enterprise infrastructure across CentOS, SUSE Linux, and VMware ESXi supporting high-traffic web and application platforms.
- Coordinated JBoss application-server optimization with senior software engineers and performed production deployments, configuration changes, and rollbacks.
- Performed JMeter load testing of critical site workflows, including shopping-cart transactions, to identify performance bottlenecks and validate capacity.
- Administered Apache web infrastructure including SSL/TLS configuration, and automated JBoss and Apache deployments with shell scripting alongside centralized monitoring.
- Configured and supported Citrix NetScaler load balancing, including virtual servers, service groups, health monitoring, and traffic distribution.

Network Administrator — ND Industries, Inc.

2005 – 2009 | Troy, MI

- Supported a carrier-managed AT&T MPLS WAN using OSPF across multiple satellite offices; troubleshot routing and connectivity on Cisco IOS in coordination with AT&T.
- Built and maintained OpenVPN remote access authenticated against RADIUS and LDAP.
- Operated core Linux infrastructure on Gentoo and Ubuntu, including qmail/Courier-IMAP mail, BIND DNS, Apache hosting, Samba file services, MySQL, and Linux RAID storage.
- Implemented company-wide Nagios monitoring and documented systems and projects using MediaWiki.
- Co-engineered a low-cost storage array using ATA-over-Ethernet blades managed with Linux mdadm.

Earlier career: Systems Analyst and Systems Administrator roles with Ford Motor Company (via Ciber/NTT), Excite@Home, ClickAction (Yesmail), and regional ISPs.